

Exercise 9

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Exercise 3.8(E.3.2, page 171)

Derive Eq.3.3.15

$$G_E^R(x, y; x', y') = \sum_m -\frac{i}{\hbar v_m} \chi_m(y) \chi_m(y') e^{ik_m |x-x'|}$$

where $k_m = \frac{\sqrt{2m(E-\epsilon_{m,0})}}{\hbar}$ and $v_m = \frac{\hbar k}{m}$. For the Green's function of an infinite wire using an eigenfunction expansion Eq.3.3.17.

$$G^R(\mathbf{r}, \mathbf{r}') = \sum_{\alpha} \frac{\psi_{\alpha}(\mathbf{r}) \psi_{\alpha}^*(\mathbf{r}')}{E - \epsilon_{\alpha} + i\eta}$$

Exercise 3.9(E.3.3, page 171)

Show that the Green's function for a semi-infinite lead p can be written as

$$g_p^R(p_i, p_j) = -\frac{1}{t} \sum_m \chi_m(p_i) \exp(+ik_m a) \chi_m(p_j)$$

where the index m represents the propagating modes in lead p .